

# OLED (Displaying a character)

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- Each character is represented as a set of bytes forming a **bitmap**.
- There are 2 different font styles 5x7 and 6x8
- Let's say we use **6x8 font**: 6 columns wide and 8 rows high.
- OLED sends data column by column, and each column is one byte.

Each ● is a pixel ON

Each ○ is a pixel OFF

Column numbers → 0 1 2 3 4 5

|       |   |   |   |   |   |   |
|-------|---|---|---|---|---|---|
| Row 0 | ○ | ● | ● | ● | ● | ○ |
| Row 1 | ● | ○ | ○ | ○ | ○ | ● |
| Row 2 | ● | ○ | ○ | ○ | ○ | ● |
| Row 3 | ● | ● | ● | ● | ● | ● |
| Row 4 | ● | ○ | ○ | ○ | ○ | ● |
| Row 5 | ● | ○ | ○ | ○ | ○ | ● |
| Row 6 | ● | ○ | ○ | ○ | ○ | ● |
| Row 7 | ○ | ○ | ○ | ○ | ○ | ○ |

- So according to the pattern,  
Column 0 has :  
Row 0 = 0  
Row 1 = 1  
Row 2 = 1  
Row 3 = 1  
Row 4 = 1  
Row 5 = 1  
Row 6 = 1  
Row 7 = 0  
In Binary val is 01111110, in hexadecimal, 0x7E  
Similarly, col 1,2,3,4 : 0x09, col 5: 0x7E
- These values are stored in 2D array, each row contains the value of one character, each character has 6 bytes

